

# Assignment 1

## Part 1

Let  $f(x) = x^2 + 5$ , so  $f'(x) = 2x$ . Points are  $(x_0, y_0)$ :

$$(0, 0), (-4, 0), (-8, 0), (2, 0), (6, 0).$$

The point  $x$  that minimizes the distance will also minimize the equation:

$$D(x) = (x - x_0)^2 + (f(x) - y_0)^2.$$

Setting the derivative to zero and simplifying:

$$\begin{aligned} 0 &= 2(x - x_0) + 2(f(x) - y_0)f'(x) \\ 0 &= (x - x_0) + (2x^3 + 10x - 2xy_0) \end{aligned}$$

## Analytical solution

a)  $x_0 = 0, y_0 = 0$

$$\begin{aligned} 0 &= x + 2x^3 + 10x \\ 0 &= 2x^3 + 11x \\ 0 &= 2x(x^2 + 11) \\ \therefore x &= 0 \end{aligned}$$

$$f(x) = 5, \quad d = \sqrt{(x - x_0)^2 + (f(x) - y_0)^2} = \sqrt{25} = 5$$

b)  $x_0 = -4, y_0 = 0$

$$\begin{aligned} 0 &= (x + 4) + 2x^3 + 10x \\ 0 &= 2x^3 + 11x + 4 \\ x &\approx -0.35 \end{aligned}$$

$$f(x) = 5.1225, \quad d = \sqrt{(-0.35 + 4)^2 + (5.1225)^2}, \quad d = 6.290$$

c)  $x_0 = -8, y_0 = 0$

$$\begin{aligned} 0 &= (x + 8) + 2x^3 + 10x \\ 0 &= 2x^3 + 11x + 8 \\ x &\approx -0.67 \end{aligned}$$

$$f(x) = 5.449, \quad d = \sqrt{(-0.67 + 8)^2 + (5.449)^2}, \quad d = 9.133$$

d)  $x_0 = 2, y_0 = 0$

$$0 = (x - 2) + 2x^3 + 10x$$

$$0 = 2x^3 + 11x - 2$$

$$x \approx 0.18$$

$$f(x) = 5.03, \quad d = \sqrt{(0.18 - 2)^2 + (5.03)^2}, \quad d = 5.349$$

e)  $x_0 = 6, y_0 = 0$

$$0 = (x - 6) + 2x^3 + 10x$$

$$0 = 2x^3 + 11x - 6$$

$$x \approx 0.52$$

$$f(x) = 5.27, \quad d = \sqrt{(0.52 - 6)^2 + (5.27)^2}, \quad d = 7.603$$

## Newton-Raphson

Calculating derivatives and simplifying:

$$D(x) = (x - x_0)^2 + (f(x) - y_0)^2$$

$$D'(x) = 2(x - x_0) + 2(f(x) - y_0)f'(x)$$

$$D'(x) = 2(2x^3 + 11x - 2xy_0 - x_0)$$

$$D''(x) = 2(6x^2 + 11 - 2y_0)$$

a)  $x_0 = 0, y_0 = 0$ , initial  $x = 0$

$$D'(x) = 4x^3 + 22x$$

$$D''(x) = 12x^2 + 22$$

$$x_1 = x - \frac{4x^3 + 22x}{12x^2 + 22}$$

$$x_1 = 0$$

$$x_0 = x_1$$

$$\therefore x = 0, \quad f(x) = 5, \quad d = 5$$

b)  $x_0 = -4, y_0 = 0$

$$D'(x) = 4x^3 + 22x + 8$$

$$D''(x) = 12x^2 + 22$$

$$x_1 = x - \frac{4x^3 + 22x + 8}{12x^2 + 22}$$

$$x_1 = \frac{-8}{22}$$

$$x_1 = -0.363636$$

$$x_2 = x_1 - \frac{4x_1^3 + 22x_1 + 8}{12x_1^2 + 22}$$

$$x_2 = \frac{-8}{22} - \frac{-0.1923}{23.5868}$$

$$x_2 = \frac{-8}{22} + 0.008153$$

$$x_2 = -0.355483$$

c)  $x_0 = -8, y_0 = 0$

$$D'(x) = 4x^3 + 22x + 16$$

$$D''(x) = 12x^2 + 22$$

$$x_1 = x - \frac{4x^3 + 22x + 16}{12x^2 + 22}$$

$$x_1 = \frac{-16}{22}$$

$$x_1 = -0.727273$$

$$x_2 = x_1 - \frac{4x_1^3 + 22x_1 + 16}{12x_1^2 + 22}$$

$$x_2 = \frac{-16}{22} - \frac{-1.5387}{28.347}$$

$$x_2 = \frac{-16}{22} + 0.054281$$

$$x_2 = -0.672992$$

d)  $x_0 = 2, y_0 = 0$

$$D'(x) = 4x^3 + 22x - 4$$

$$D''(x) = 12x^2 + 22$$

$$x_1 = x - \frac{4x^3 + 22x - 4}{12x^2 + 22}$$

$$x_1 = \frac{4}{22}$$

$$x_1 = 0.181818$$

$$x_2 = x_1 - \frac{4x_1^3 + 22x_1 - 4}{12x_1^2 + 22}$$

$$x_2 = \frac{4}{22} - \frac{0.002364}{22.39670}$$

$$x_2 = \frac{4}{22} - 0.00010555$$

$$x_2 = 0.18171$$

e)  $x_0 = 6, y_0 = 0$

$$D'(x) = 4x^3 + 22x - 12$$

$$D''(x) = 12x^2 + 22$$

$$x_1 = x - \frac{4x^3 + 22x - 12}{12x^2 + 22}$$

$$x_1 = \frac{12}{22}$$

$$x_1 = 0.545454$$

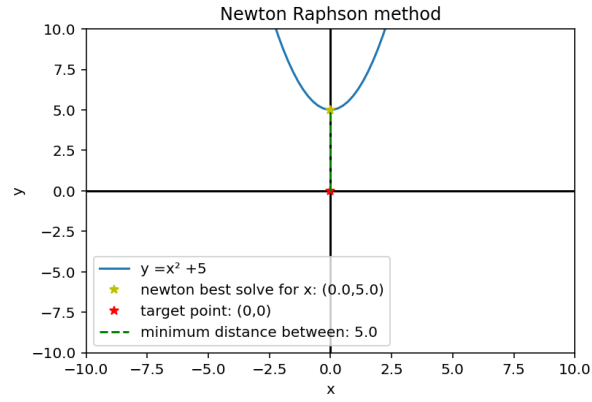
$$x_2 = x_1 - \frac{4x_1^3 + 22x_1 - 12}{12x_1^2 + 22}$$

$$x_2 = \frac{12}{22} - \frac{0.64914}{25.5702}$$

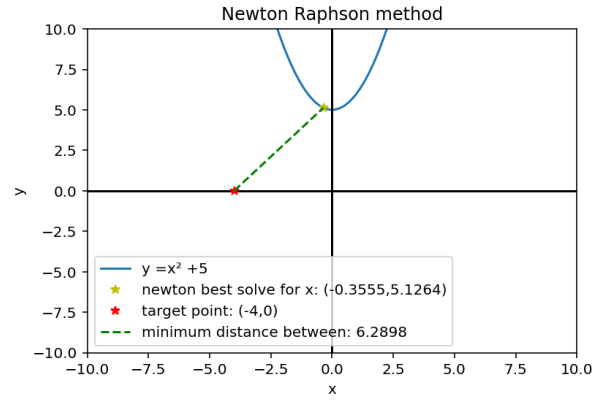
$$x_2 = \frac{12}{22} - 0.02539$$

$$x_2 = 0.52006$$

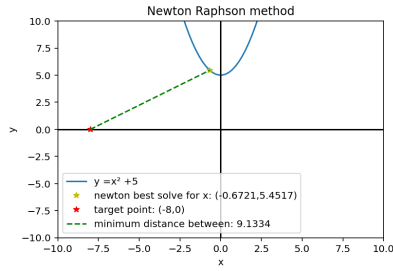
## Newton-Raphson Plotting



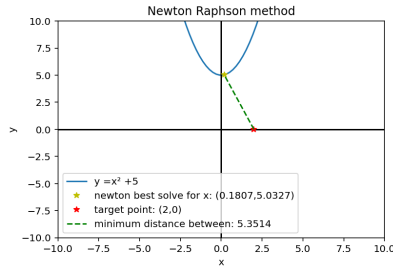
(a)  $x_0 = 0, y_0 = 0$



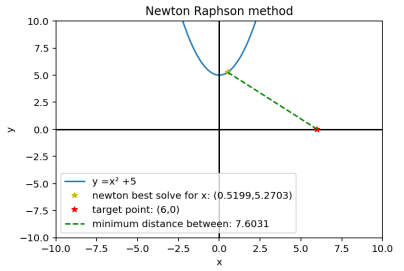
(b)  $x_0 = -4, y_0 = 0$



(c)  $x_0 = -8, y_0 = 0$



(d)  $x_0 = 2, y_0 = 0$



(e)  $x_0 = 6, y_0 = 0$

Figure 1: Newton Raphson plotted outputs.

Table 1: Data for Newton Raphson search

| Points              | Iterations | X progress                                                                                          |
|---------------------|------------|-----------------------------------------------------------------------------------------------------|
| $x_0 = 0, y_0 = 0$  | 1          | 0.0                                                                                                 |
| $x_0 = -4, y_0 = 0$ | 4          | -0.363636363636364, -0.355481939224056,<br>-0.355469692902039, -0.355469692874835                   |
| $x_0 = -8, y_0 = 0$ | 5          | -0.727272727272727, -0.67299231380864, -0.672078372679461,<br>-0.67207812677897, -0.672078126778952 |
| $x_0 = 2, y_0 = 0$  | 4          | 0.181818181818182, 0.180744716538074,<br>0.180744604479351, 0.18074460447935                        |
| $x_0 = 6, y_0 = 0$  | 4          | 0.545454545454545, 0.520068167126991,<br>0.519903667723888, 0.519903661034705                       |

## Part 2

### Golden section search on $D(x)$

Choosing  $a = -1, b = 1$  for initial search:

$$\tau = \frac{1 + \sqrt{5}}{2} = 1.618$$

$$x_1 = a + (2 - \tau)(b - a)$$

$$x_1 = -0.236$$

$$x_2 = b - (2 - \tau)(b - a)$$

$$x_2 = 0.236$$

$$D(x) = (x - x_0)^2 + (f(x) - y_0)^2, \quad f(x) = x^2 + 5$$

$$f(x_1) = 5.056, \quad f(x_2) = 5.056$$

a)  $x_0 = 0, y_0 = 0$

$$D(x_1) = (x_1)^2 + (f(x_1))^2 = 25.61$$

$$D(x_2) = 25.618$$

$$D(x_1) = D(x_2)$$

$$\therefore x = 0, \quad f(x) = 5$$

b)  $x_0 = -4, y_0 = 0$

$$D(x_1) = (0.236 + 4)^2 + f(x_1)^2 = 39.73$$

$$D(x_2) = 43.51$$

$$D(x_1) < D(x_2)$$

$$b = 0.236$$

$$x_2 = -0.236$$

$$x_1 = a + (2 - \tau)(b - a)$$

$$x_1 = -0.528$$

After 1 iteration best  $x = -0.382$ ,  $f(x) = 5.146$

c)  $x_0 = -8$ ,  $y_0 = 0$

$$D(x_1) = (-0.236 + 8)^2 + f(x_1)^2 = 85.88$$

$$D(x_2) = 93.39$$

$$D(x_1) < D(x_2)$$

$$b = 0.236$$

$$x_2 = -0.236$$

$$x_1 = a + (2 - \tau)(b - a)$$

$$x_1 = -0.528$$

After 1 iteration best  $x = -0.382$ ,  $f(x) = 5.146$

d)  $x_0 = 2$ ,  $y_0 = 0$

$$D(x_1) = (-0.236 - 2)^2 + f(x_1)^2 = 30.56$$

$$D(x_2) = 28.67$$

$$D(x_1) > D(x_2)$$

$$a = -0.236$$

$$x_1 = 0.236$$

$$x_2 = b - (2 - \tau)(b - a)$$

$$x_2 = 0.528$$

After 1 iteration best  $x = 0.382$ ,  $f(x) = 5.146$

e)  $x_0 = 6$ ,  $y_0 = 0$

$$D(x_1) = (-0.236 - 6)^2 + f(x_1)^2 = 64.45$$

$$D(x_2) = 58.79$$

$$D(x_1) > D(x_2)$$

$$a = -0.236$$

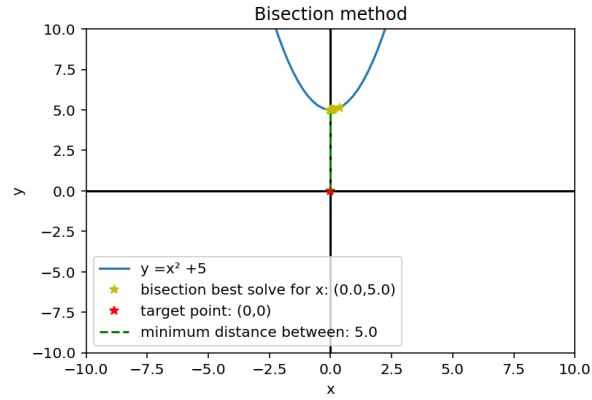
$$x_1 = 0.236$$

$$x_2 = b - (2 - \tau)(b - a)$$

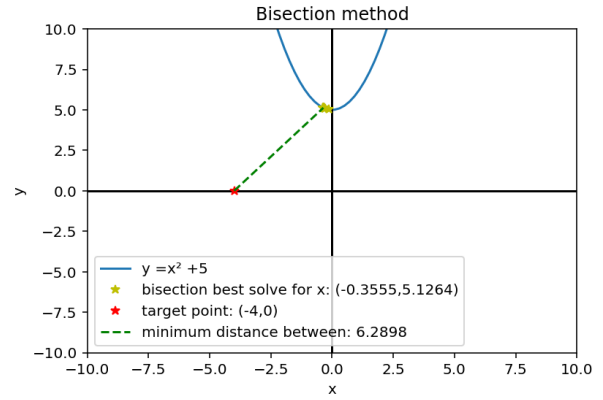
$$x_2 = 0.528$$

After 1 iteration best  $x = 0.382$ ,  $f(x) = 5.146$

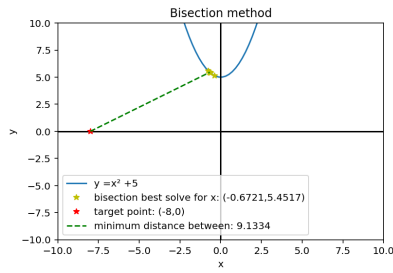
## Bisection Plotting



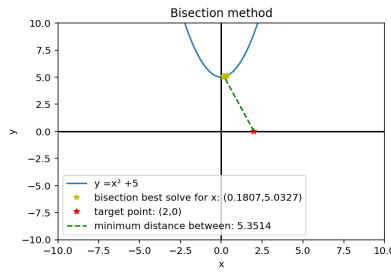
(a)  $x_0 = 0, y_0 = 0$



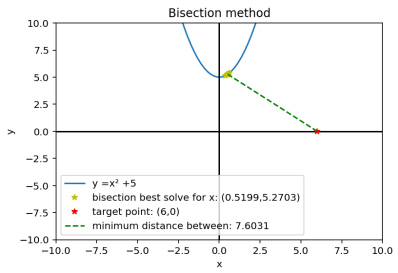
(b)  $x_0 = -4, y_0 = 0$



(c)  $x_0 = -8, y_0 = 0$



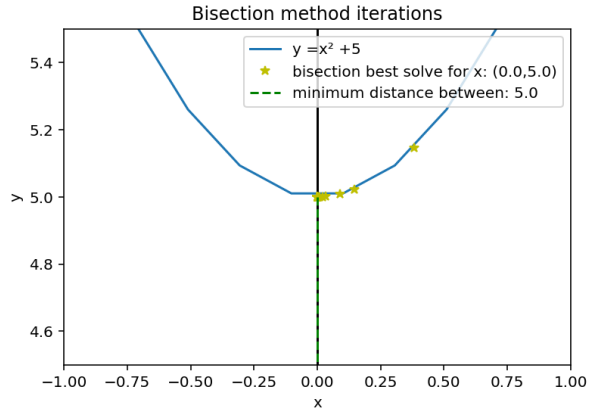
(d)  $x_0 = 2, y_0 = 0$



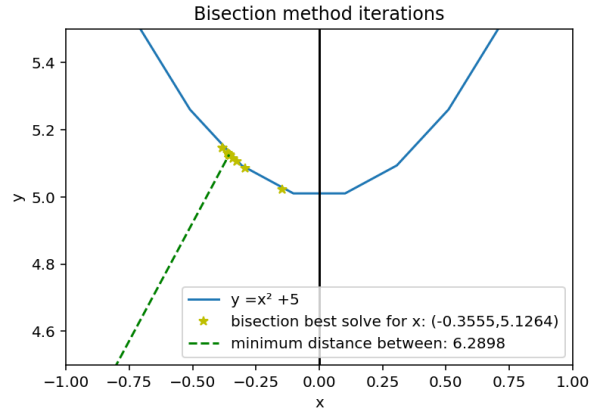
(e)  $x_0 = 6, y_0 = 0$

Figure 2: Bisection plotted outputs.

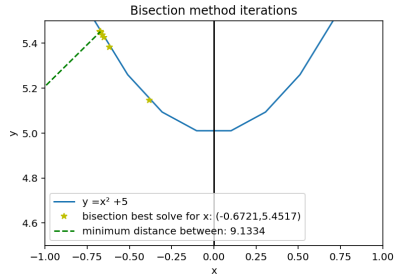




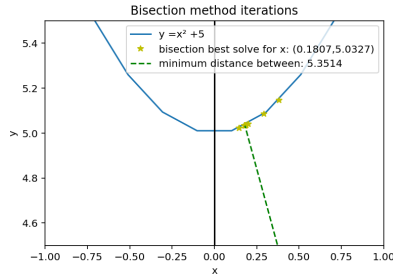
(a)  $x_0 = 0, y_0 = 0$



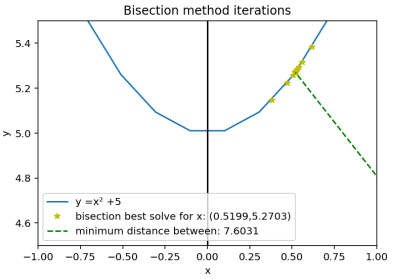
(b)  $x_0 = -4, y_0 = 0$



(c)  $x_0 = -8, y_0 = 0$



(d)  $x_0 = 2, y_0 = 0$



(e)  $x_0 = 6, y_0 = 0$

Figure 3: Bisection iterations, Number of iterations = 35

## Fitting to an Equation

Data points: (0, 0.5), (2, 3.3), (1, 1.5), (3, 7.5).

Mean squared error:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

## Analytical line

$$\frac{1}{4} \sum_{i=1}^4 (y - (mx + b))^2 = \frac{1}{4} \sum_{i=1}^4 (y^2 - 2y(mx + b) + m^2x^2 + 2mxb + b^2)$$

$$\frac{\partial}{\partial m} \rightarrow 0 = \frac{1}{4} \sum_{i=1}^4 \left( -2 \underbrace{yx}_{=31} + 2m \underbrace{x^2}_{=14} + 2b \underbrace{x}_{=6} \right)$$

$$0 = \frac{2}{4} (-31 + 14m + 6b)$$

$$b = \frac{31}{6} - \frac{14m}{6}$$

$$\frac{\partial}{\partial b} \rightarrow 0 = \frac{1}{4} \sum_{i=1}^4 \left( -2 \underbrace{y}_{=13} + 2m \underbrace{x}_{=6} + 2 \underbrace{b}_{=4} \right)$$

$$0 = \frac{2}{4} (-13 + 6m + 4b)$$

$$b = \frac{13}{4} - \frac{6m}{4}$$

$$\frac{13}{4} - \frac{6m}{4} = \frac{31}{6} - \frac{14m}{6} \implies m = 2.3, b = -0.2$$

$$y = 2.3x - 0.2$$

$$\text{MSE} = \frac{1}{4} \sum_{i=1}^4 (y_i - 2.3x_i + 0.2)^2$$

$$\text{MSE} = \frac{1}{4} (0.7^2 - 0.9^2 - 0.6^2 + 0.8^2)$$

$$\text{MSE} = 0.575$$

# Newton–Raphson Multivariate line

$$X_1 = X_0 - H^{-1}g$$

$$f(x) = \frac{1}{4} \sum_{i=1}^4 (y - mx - b)^2$$

$$\frac{\partial f}{\partial m} = \frac{1}{4} \sum_{i=1}^4 (-2yx + 2mx^2 + 2xb)$$

$$\frac{\partial^2 f}{\partial m^2} = \frac{1}{4} \sum 2x^2$$

$$\frac{\partial f}{\partial b} = \frac{1}{4} \sum_{i=1}^4 (-2y + 2mx + 2b)$$

$$\frac{\partial^2 f}{\partial b^2} = \frac{1}{4} \sum 2$$

$$\frac{\partial^2 f}{\partial m \partial b} = \frac{1}{4} \sum 2x$$

$$g = \begin{bmatrix} \frac{\partial f}{\partial m} \\ \frac{\partial f}{\partial b} \end{bmatrix}, \quad H = \begin{bmatrix} \frac{\partial^2 f}{\partial m^2} & \frac{\partial^2 f}{\partial b \partial m} \\ \frac{\partial^2 f}{\partial m \partial b} & \frac{\partial^2 f}{\partial b^2} \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 3 & 2 \end{bmatrix}$$

$$\det H = 14 - 9 = 5, \quad H^{-1} = \frac{1}{5} \begin{bmatrix} 2 & -3 \\ -3 & 7 \end{bmatrix}$$

Guess  $m = 1$ ,  $b = 1$ :

$$g = \begin{bmatrix} \frac{2}{4}(-31 + 14m + 6b) \\ \frac{2}{4}(-13 + 6m + 4b) \end{bmatrix} = \begin{bmatrix} -5.5 \\ -1.5 \end{bmatrix}$$

$$H^{-1}g = \frac{1}{5} \begin{bmatrix} 2 & -3 \\ -3 & 7 \end{bmatrix} \begin{bmatrix} -5.5 \\ -1.5 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} -11 + 4.5 \\ 16.5 - 10.5 \end{bmatrix} = \begin{bmatrix} -1.3 \\ 1.2 \end{bmatrix}$$

$$X_1 = (1, 1) - (-1.3, 1.2) = (2.3, -0.2)$$

## Analytical parabola

$$\frac{1}{4} \sum_{i=1}^4 (y - (ax^2 + bx + c))^2$$

$$\frac{\partial}{\partial a} \rightarrow 0 = \frac{1}{4} \sum_{i=1}^4 \left( -2 \underbrace{yx^2}_{=83} + 2a \underbrace{x^4}_{=98} + 2b \underbrace{x^3}_{=36} + 2c \underbrace{x^2}_{=14} \right)$$

$$\textcircled{1} \quad 0 = \frac{2}{4} (-83 + 98a + 36b + 14c)$$

$$\frac{\partial}{\partial b} \rightarrow 0 = \frac{1}{4} \sum_{i=1}^4 \left( -2 \underbrace{yx}_{=31} + 2a \underbrace{x^3}_{=36} + 2b \underbrace{x^2}_{=14} + 2c \underbrace{x}_{=6} \right)$$

$$\textcircled{2} \quad 0 = \frac{2}{4} (-31 + 36a + 14b + 6c)$$

$$\frac{\partial}{\partial c} \rightarrow 0 = \frac{1}{4} \sum_{i=1}^4 \left( -2 \underbrace{y}_{=13} + 2a \underbrace{x^2}_{=14} + 2b \underbrace{x}_{=6} + 2 \underbrace{c}_{=4} \right)$$

$$\textcircled{3} \quad 0 = \frac{2}{4} (-13 + 14a + 6b + 4c)$$

$$a = 0.75, \quad b = 0.05, \quad c = 0.55$$

$$y = 0.75x^2 + 0.05x + 0.55$$

$$\text{MSE} = \frac{1}{4} \sum_{i=1}^4 (y_i - 0.75x_i^2 - 0.05x_i - 0.55)^2$$

$$\text{MSE} = \frac{1}{4} (-0.05^2 - 0.15^2 + 0.15^2 + 0.05^2)$$

$$\text{MSE} = 0.0125$$

## Newton–Raphson Multivariate parabola

$$g = \begin{bmatrix} \frac{\partial f}{\partial a} \\ \frac{\partial f}{\partial b} \\ \frac{\partial f}{\partial c} \end{bmatrix}, \quad H = \frac{1}{2} \begin{bmatrix} \sum x^4 & \sum x^3 & \sum x^2 \\ \sum x^3 & \sum x^2 & \sum x \\ \sum x^2 & \sum x & \sum 1 \end{bmatrix}$$

$$H = \begin{bmatrix} 49 & 18 & 7 \\ 18 & 7 & 3 \\ 7 & 3 & 2 \end{bmatrix}, \quad H^{-1} = \frac{1}{10} \begin{bmatrix} 5 & -15 & 5 \\ -15 & 49 & -21 \\ 5 & -21 & 19 \end{bmatrix}$$

Guess  $a = 0$ ,  $b = 0$ ,  $c = 0$ :

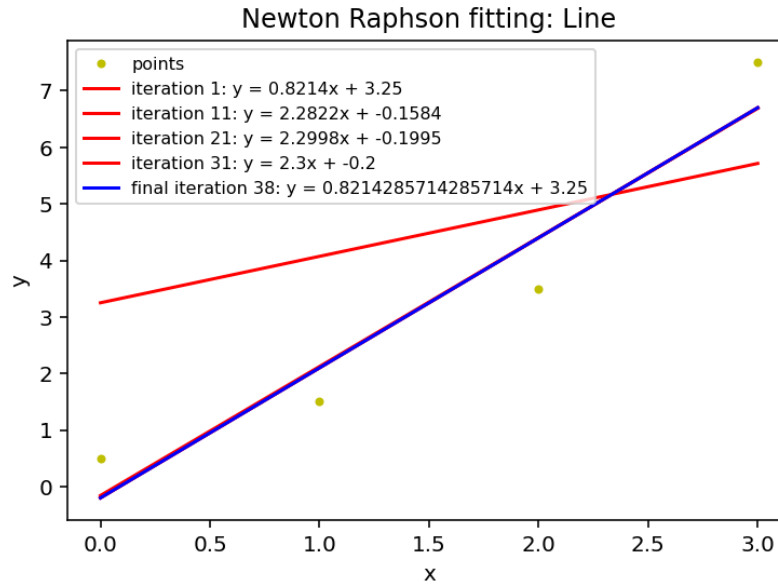
$$g = \begin{bmatrix} \frac{2}{4}(-83 + 98a + 36b + 14c) \\ \frac{2}{4}(-31 + 36a + 14b + 6c) \\ \frac{2}{4}(-13 + 14a + 6b + 4c) \end{bmatrix} = \begin{bmatrix} -41.5 \\ -15.5 \\ -6.5 \end{bmatrix}$$

$$H^{-1}g = (-0.75, -0.05, -0.55)$$

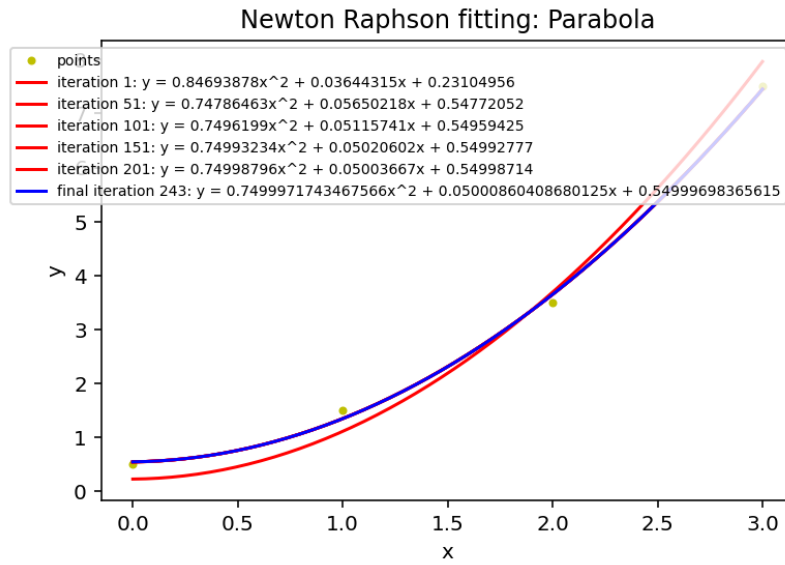
$$X_1 = (0, 0, 0) - (-0.75, -0.05, -0.55)$$

$$X_1 = (0.75, 0.05, 0.55)$$

## Newton Raphson Iterative Plotting



(a) Initial guesses:  $m = 0$ ,  $b = 0$



(b) Initial guesses:  $a_0 = 0$ ,  $b_0 = 0$ ,  $c_0 = 0$

Figure 4: Newton Raphson Step-wise method plotted.

## Code

## Part 1

### Newton Raphson

```
1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  def find_distance_newton(x0, y0, f, df, ddf, initial_guess, tolerance, max_iter):
5      x = initial_guess
6      x_progress = []
7      for i in range(max_iter):
8          #  $D'(x)$ 
9          D_prime = 2 * (x - x0) + 2 * (f(x) - y0) * df(x)
10         #  $D''(x)$ 
11         D_double_prime = 2 + 2 * (df(x)**2) + 2 * (f(x) - y0) * ddf(x)
12         # Newton-Raphson update step
13         next_x = x - D_prime / D_double_prime
14         x_progress.append(round(next_x,15))
15         if abs(next_x - x) < tolerance:
16             break
17         x = next_x
18         shortest_distance = ((x - x0)**2 + (f(x) - y0)**2)**0.5
19         return (x,shortest_distance,x_progress)
20
21  fx=lambda x : (x**2) + 5
22  dfx=lambda x : 2*x
23  ddfx=lambda x : 2
24
25
26  if __name__=="__main__":
27      #testing
28      x_range = np.linspace(-5,5)
29      y_range = fx(x_range)
30      x0_points = [0,-4,-8,2,6]
31      for i in range(5):
32          test = find_distance_newton(x0_points[i],0,fx,dfx,ddfx,0.0,1e-7,100)
33          x_guesses = np.array(test[2])
34          y_guesses = fx(x_guesses)
35          print(str(test[2]))
36          plt.figure()
37          plt.title('Newton Raphson method')
38          plt.plot(x_range,y_range, label= 'y =x\u00b2 +5')
39          plt.xlim(-10, 10)
40          plt.ylim(-10, 10)
41          plt.axvline(x=0,color='k')
42          plt.axhline(y=0,color='k')
43          plt.plot(test[0],fx(test[0]),'y*', label= 'newton best solve for x: ('+
44              ↵ str(round(test[0],4)) + ',' + str(round(fx(test[0]),4))+')')
```

```

44     plt.plot(x0_points[i],0,'r*',label='target point: ('+ str(x0_points[i]) +
    ↪      ',0)')
45     plt.plot([x0_points[i],test[0]],[0,fx(test[0])],'g--',label='minimum
    ↪      distance between: ' + str(round(test[1],4)))
46     plt.legend()
47     plt.xlabel('x')
48     plt.ylabel('y')
49     plt.show()

```

## Golden Section Search

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  def newton_raphson_line(x, y, m_guess, b_guess,tolerance):
5      b_new = b_guess
6      m_new = m_guess
7      b_current = b_new + 2*tolerance
8      m_current = m_new + 2*tolerance
9      iterations = 0
10     b_progress = []
11     m_progress = []
12     b_progress.append(b_new)
13     m_progress.append(m_new)
14     while (abs(b_new - b_current) > tolerance) & (abs(m_new - m_current) >
    ↪      tolerance):
15         b_current = b_new
16         m_current = m_new
17         b_new = (b_current) - dfdb(m_current,b_current,x,y)/df2db2(x)
18         m_new = (m_current) - dfdm(m_current,b_new,x,y)/df2dm2(x)
19         iterations+= 1
20         b_progress.append(b_new)
21         m_progress.append(m_new)
22     return (b_new, m_new,iterations,b_progress,m_progress)
23
24
25 def dfdm(m,b,x,y):
26     df = 0
27     for i in range(len(x)):
28         df += -2*y[i]*x[i]+2*m*(x[i]**2)+2*x[i]*b
29     df=df/4
30     return(df)
31
32 def dfdb(m,b,x,y):
33     df = 0
34     for i in range(len(x)):
35         df += -2*y[i]+2*m*x[i]+2*b
36     df=df/4

```



```

37     return(df)
38
39 def df2dm2(x):
40     df = 0
41     for i in range(len(x)):
42         df += 2*(x[i]**2)
43     df=df/4
44     return(df)
45
46 def df2db2(x):
47     df = 0
48     for i in range(len(x)):
49         df += 2
50     df=df/4
51     return(df)
52
53
54 if __name__=="__main__":
55     x = np.array([0,2,1,3])
56     y = np.array([0.5,3.5,1.5,7.5])
57     test_1 = newton_raphson_line(x,y,0,0,1e-7)
58     print(test_1[0],test_1[1])
59     b_progress = test_1[3]
60     m_progress = test_1[4]
61     x_range = np.linspace(0,3)
62     y_range = np.linspace(0,8)
63     plt.figure()
64     plt.title('Newton Raphson fitting: Line')
65     plt.plot(x,y, 'y.',label= 'points')
66     for i in range(1,len(b_progress),10):
67         plt.plot(x_range,x_range*m_progress[i] + b_progress[i] ,
68                 ↪ color='red',label= f'iteration {i}: y = {round(m_progress[i],4)}x +
69                 ↪ {round(b_progress[i],4)}')
70     plt.plot(x_range,x_range*test_1[1] + test_1[0] , color='blue',label=f'final
71     ↪ iteration {test_1[2]}: y = {m_progress[1]}x + {b_progress[1]}')
72     plt.legend(fontsize=8)
73     plt.xlabel('x')
74     plt.ylabel('y')
75     plt.show()

```

## Part 2

### Line Solve Iterative

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def newton_raphson_line(x, y, m_guess, b_guess,tolerance):

```

```

5     b_new = b_guess
6     m_new = m_guess
7     b_current = b_new + 2*tolerance
8     m_current = m_new + 2*tolerance
9     iterations = 0
10    b_progress = []
11    m_progress = []
12    b_progress.append(b_new)
13    m_progress.append(m_new)
14    while (abs(b_new - b_current) > tolerance) & (abs(m_new - m_current) >
↪     tolerance):
15        b_current = b_new
16        m_current = m_new
17        b_new = (b_current) - dfdb(m_current,b_current,x,y)/df2db2(x)
18        m_new = (m_current) - dfdm(m_current,b_new,x,y)/df2dm2(x)
19        iterations+= 1
20        b_progress.append(b_new)
21        m_progress.append(m_new)
22    return (b_new, m_new,iterations,b_progress,m_progress)
23
24
25    def dfdm(m,b,x,y):
26        df = 0
27        for i in range(len(x)):
28            df += -2*y[i]*x[i]+2*m*(x[i]**2)+2*x[i]*b
29        df=df/4
30        return(df)
31
32    def dfdb(m,b,x,y):
33        df = 0
34        for i in range(len(x)):
35            df += -2*y[i]+2*m*x[i]+2*b
36        df=df/4
37        return(df)
38
39    def df2dm2(x):
40        df = 0
41        for i in range(len(x)):
42            df += 2*(x[i]**2)
43        df=df/4
44        return(df)
45
46    def df2db2(x):
47        df = 0
48        for i in range(len(x)):
49            df += 2
50        df=df/4
51        return(df)

```

```

52
53
54 if __name__=="__main__":
55     x = np.array([0,2,1,3])
56     y = np.array([0.5,3.5,1.5,7.5])
57     test_1 = newton_raphson_line(x,y,0,0,1e-7)
58     print(test_1[0],test_1[1])
59     b_progress = test_1[3]
60     m_progress = test_1[4]
61     x_range = np.linspace(0,3)
62     y_range = np.linspace(0,8)
63     plt.figure()
64     plt.title('Newton Raphson fitting: Line')
65     plt.plot(x,y, 'y.',label= 'points')
66     for i in range(1,len(b_progress),10):
67         plt.plot(x_range,x_range*m_progress[i] + b_progress[i] ,
68                 ↪ color='red',label= f'iteration {i}: y = {round(m_progress[i],4)}x +
69                 ↪ {round(b_progress[i],4)}')
70     plt.plot(x_range,x_range*test_1[1] + test_1[0] , color='blue',label=f'final
71     ↪ iteration {test_1[2]}: y = {m_progress[1]}x + {b_progress[1]}')
72     plt.legend(fontsize=8)
73     plt.xlabel('x')
74     plt.ylabel('y')
75     plt.show()

```

## Parabola Solve Iterative

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  def newton_raphson_parabola(x, y, a_guess, b_guess, c_guess,tolerance):
5      a_new = a_guess
6      b_new = b_guess
7      c_new = c_guess
8      a_current = a_new + 2*tolerance
9      b_current = b_new + 2*tolerance
10     c_current = c_new + 2*tolerance
11     iterations = 0
12     a_progress = []
13     b_progress = []
14     c_progress = []
15     a_progress.append(a_new)
16     b_progress.append(b_new)
17     c_progress.append(c_new)
18     while (abs(a_new - a_current) > tolerance) & (abs(b_new - b_current) >
19     ↪ tolerance)& (abs(c_new - c_current) > tolerance):
20         a_current = a_new
21         b_current = b_new

```

```

21         c_current = c_new
22         a_new = (a_current) - dfda(a_current,b_current,c_current,x,y)/df2da2(x)
23         b_new = (b_current) - dfdb(a_new,b_current,c_current,x,y)/df2db2(x)
24         c_new = (c_current) - dfdc(a_new,b_new,c_current,x,y)/df2dc2(x)
25         iterations+= 1
26         a_progress.append(a_new)
27         b_progress.append(b_new)
28         c_progress.append(c_new)
29     return (a_new, b_new, c_new, iterations, a_progress, b_progress, c_progress)
30
31
32 def dfda(a,b,c,x,y):
33     df = 0
34     for i in range(len(x)):
35         df += -2*y[i]*(x[i]**2)+2*a*(x[i]**4)+2*b*(x[i]**3)+2*c*(x[i]**2)
36     df=df/4
37     return(df)
38
39 def dfdb(a,b,c,x,y):
40     df = 0
41     for i in range(len(x)):
42         df += -2*y[i]*x[i]+2*a*(x[i]**3)+2*b*(x[i]**2)+2*c*x[i]
43     df=df/4
44     return(df)
45
46 def dfdc(a,b,c,x,y):
47     df = 0
48     for i in range(len(x)):
49         df += -2*y[i]+2*a*(x[i]**2)+2*b*(x[i])+2*c
50     df=df/4
51     return(df)
52
53 def df2da2(x):
54     df = 0
55     for i in range(len(x)):
56         df += 2*(x[i]**4)
57     df=df/4
58     return(df)
59
60 def df2db2(x):
61     df = 0
62     for i in range(len(x)):
63         df += 2*(x[i]**2)
64     df=df/4
65     return(df)
66
67 def df2dc2(x):
68     df = 0

```

```

69     for i in range(len(x)):
70         df += 2*(x[i]**0)
71     df=df/4
72     return(df)
73
74 if __name__=="__main__":
75     x = np.array([0,2,1,3])
76     y = np.array([0.5,3.5,1.5,7.5])
77     test_1 = newton_raphson_parabola(x,y,0,0,0,1e-7)
78     print(test_1[0],test_1[1], test_1[2], test_1[3])
79     a_progress = test_1[4]
80     b_progress = test_1[5]
81     c_progress = test_1[6]
82     x_range = np.linspace(0,3)
83     y_range = np.linspace(0,8)
84     plt.figure()
85     plt.title('Newton Raphson fitting: Parabola')
86     plt.plot(x,y, 'y.',label= 'points')
87     for i in range(1,len(c_progress),50):
88         plt.plot(x_range,a_progress[i]*(x_range**2) + x_range*b_progress[i] +
89             ↪ c_progress[i], color='red',label= f'iteration {i}: y =
90             ↪ {round(a_progress[i],8)}x^2 + {round(b_progress[i],8)}x +
91             ↪ {round(c_progress[i],8)}')
92     plt.plot(x_range,test_1[0]*(x_range**2) + x_range*test_1[1] + test_1[2],
93         ↪ color='blue',label=f'final iteration {test_1[3]}: y = {test_1[0]}x^2 +
94         ↪ {test_1[1]}x + {test_1[2]}')
95     plt.legend(fontsize=7)
96     plt.xlabel('x')
97     plt.ylabel('y')
98     plt.show()

```

## Line Solve Multivariate

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  def newton_raphson_line(x, y, m_guess, b_guess):
5      H = np.array([[df2dm2(x),df2dmdb(x)],[df2dmdb(x),df2db2(x)]])
6      Hinv = np.linalg.inv(H)
7      g = np.array([dfdmd(m_guess,b_guess,x,y),dfddb(m_guess,b_guess,x,y)])
8      gHinv = Hinv@g
9      final_guess = np.array([m_guess,b_guess]) - gHinv
10     return (final_guess)
11
12
13 def dfdm(m,b,x,y):
14     df = 0
15     for i in range(len(x)):

```

```

16         df += -2*y[i]*x[i]+2*m*(x[i]**2)+2*x[i]*b
17     df=df/4
18     return(df)
19
20 def dfdb(m,b,x,y):
21     df = 0
22     for i in range(len(x)):
23         df += -2*y[i]+2*m*x[i]+2*b
24     df=df/4
25     return(df)
26
27 def df2dm2(x):
28     df = 0
29     for i in range(len(x)):
30         df += 2*(x[i]**2)
31     df=df/4
32     return(df)
33
34 def df2db2(x):
35     df = 0
36     for i in range(len(x)):
37         df += 2
38     df=df/4
39     return(df)
40
41 def df2dmdb(x):
42     df = 0
43     for i in range(len(x)):
44         df += 2*x[i]
45     df=df/4
46     return(df)
47
48 if __name__=="__main__":
49     x = np.array([0,2,1,3])
50     y = np.array([0.5,3.5,1.5,7.5])
51     test_1 = newton_raphson_line(x,y,1,1)
52     print(test_1)

```

## Parabola Solve Multivariate

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def newton_raphson_parabola(x, y, a_guess, b_guess,c_guess):
5     H =
6         ↪ 0.5*np.array([[sum(x[:]**4),sum(x[:]**3),sum(x[:]**2)],
7                        [sum(x[:]**3),sum(x[:]**2),sum(x[:]**1)],
8                        [sum(x[:]**2),sum(x[:]**1),sum(x[:]**0)]])
9     Hinv = np.linalg.inv(H)

```

```

7      g =
      ↪ np.array([dfda(a_guess,b_guess,c_guess,x,y),dfdb(a_guess,b_guess,c_guess,x,y),dfdc(a_g
8      gHinv = Hinv@g
9      final_guess = np.array([a_guess,b_guess,c_guess]) - gHinv
10     return (final_guess)
11
12
13     def dfda(a,b,c,x,y):
14         df = 0
15         for i in range(len(x)):
16             df += -2*y[i]*(x[i]**2)+2*a*(x[i]**4)+2*b*(x[i]**3)+2*c*(x[i]**2)
17         df=df/4
18         return(df)
19
20     def dfdb(a,b,c,x,y):
21         df = 0
22         for i in range(len(x)):
23             df += -2*y[i]*x[i]+2*a*(x[i]**3)+2*b*(x[i]**2)+2*c*x[i]
24         df=df/4
25         return(df)
26
27     def dfdc(a,b,c,x,y):
28         df = 0
29         for i in range(len(x)):
30             df += -2*y[i]+2*a*(x[i]**2)+2*b*(x[i])+2*c
31         df=df/4
32         return(df)
33
34     if __name__=="__main__":
35         x = np.array([0,2,1,3])
36         y = np.array([0.5,3.5,1.5,7.5])
37         test_1 = newton_raphson_parabola(x,y,0,0,0)
38         print(test_1)

```

## Handwritten Notes

Part I

$$f(x) = x^2 + 5 \quad (0,0), (-4,0), (-8,0), (2,0), (6,0)$$

$$f'(x) = 2x$$

$$D(x) = (x - x_0)^2 + (f(x) - y_0)^2$$

$$0 = 2(x - x_0) + 2(f(x) - y_0)f'(x)$$

$$0 = (x - x_0) + (2x^3 + 10x - 2xy_0)$$

a)  $x_0 = 0 \quad y_0 = 0$

$$0 = x + 2x^3 + 10x$$

$$0 = 2x^3 + 11x$$

$$2x(x^2 + 11) = 0$$

$$\therefore x = 0$$

$$f(x) = 5$$

$$d = \sqrt{(x - x_0)^2 + (f(x) - y_0)^2}$$

$$d = \sqrt{25}$$

$$d = 5$$

b)  $x_0 = -4 \quad y_0 = 0$

$$0 = (x + 4) + 2x^3 + 10x$$

$$0 = 2x^3 + 11x + 4$$

$$x \approx -0.35$$

$$f(x) = 5.1225$$

$$d = \sqrt{(-0.35 + 4)^2 + (5.1225)^2}$$

$$d = 6.290$$

c)  $x_0 = -8 \quad y_0 = 0$

$$0 = (x + 8) + 2x^3 + 10x$$

$$0 = 2x^3 + 11x + 8$$

$$x \approx -0.67$$

$$f(x) = 5.449$$

$$d = \sqrt{(-0.67 + 8)^2 + (5.449)^2}$$

$$d = 9.133$$

d)  $x_0 = 2 \quad y_0 = 0$

$$0 = (x - 2) + 2x^3 + 10x$$

$$0 = 2x^3 + 11x - 2$$

$$x \approx 0.18$$

$$f(x) = 5.03$$

$$d = \sqrt{(0.18 - 2)^2 + (5.03)^2}$$

$$d = 5.349$$

e)  $x_0 = 6 \quad y_0 = 0$

$$0 = (x - 6) + 2x^3 + 10x$$

$$0 = 2x^3 + 11x - 6$$

$$x \approx 0.52$$

$$f(x) = 5.27$$

$$d = \sqrt{(0.52 - 6)^2 + (5.27)^2}$$

$$d = 7.603$$

Newton-Raphson

$$D(x) = (x - x_0)^2 + (f(x) - y_0)^2$$

$$D'(x) = 2(x - x_0) + 2(f(x) - y_0)f'(x)$$

$$D'(x) = 2(2x^3 + 11x - 2xy_0 - x_0)$$

$$D'(x) = 2(6x^2 + 11 - 2y_0)$$

a)  $x_0 = 0 \quad y_0 = 0 \quad \text{initial } x = 0$

$$D'(x) = 4x^2 + 22x$$

$$D''(x) = 12x + 22$$

$$x_1 = x - \frac{4x^3 + 22x}{12x^2 + 22}$$

$$x_1 = 0$$

$$x_0 = x_1$$

$$\therefore x = 0, f(x) = 5, d = 5$$

b)  $x_0 = -4 \quad y_0 = 0$

$$D'(x) = 4x^2 + 22x + 8$$

$$D''(x) = 12x + 22$$

$$x_1 = x - \frac{4x^3 + 22x + 8}{12x^2 + 22}$$

$$x_1 = -\frac{8}{22}$$

$$x_1 = -0.3636$$

$$x_2 = x_1 - \frac{4x_1^3 + 22x_1 + 8}{12x_1^2 + 22}$$

$$x_2 = -\frac{8}{22} - \frac{-0.1923}{25.5868}$$

$$x_2 = -\frac{8}{22} + 0.007153$$

$$x_2 = -0.355483$$

c)  $x_0 = -8 \quad y_0 = 0$

$$D'(x) = 4x^2 + 22x + 16$$

$$D''(x) = 12x + 22$$

$$x_1 = x - \frac{4x^3 + 22x + 16}{12x^2 + 22}$$

$$x_1 = -\frac{16}{22}$$

$$x_1 = -0.727273$$

$$x_2 = x_1 - \frac{4x_1^3 + 22x_1 + 16}{12x_1^2 + 22}$$

$$x_2 = -\frac{16}{22} - \frac{-1.5287}{28.347}$$

$$x_2 = -\frac{16}{22} + 0.054281$$

$$x_2 = -0.672992$$

d)  $x_0 = 2 \quad y_0 = 0$

$$D'(x) = 4x^2 + 22x - 4$$

$$D''(x) = 12x + 22$$

$$x_1 = x - \frac{4x^3 + 22x - 4}{12x^2 + 22}$$

$$x_1 = \frac{4}{22}$$

$$x_1 = 0.181818$$

$$x_2 = x_1 - \frac{4x_1^3 + 22x_1 - 4}{12x_1^2 + 22}$$

$$x_2 = \frac{4}{22} - \frac{0.003361}{22.39670}$$

$$x_2 = \frac{4}{22} - 0.00010555$$

$$x_2 = 0.18171$$

e)  $x_0 = 6 \quad y_0 = 0$

$$D'(x) = 4x^2 + 22x - 12$$

$$D''(x) = 12x + 22$$

$$x_1 = x - \frac{4x^3 + 22x - 12}{12x^2 + 22}$$

$$x_1 = \frac{12}{22}$$

$$x_1 = 0.545454$$

$$x_2 = x_1 - \frac{4x_1^3 + 22x_1 - 12}{12x_1^2 + 22}$$

$$x_2 = \frac{12}{22} - \frac{0.64914}{25.5702}$$

$$x_2 = \frac{12}{22} - 0.02539$$

$$x_2 = 0.52006$$



$$\begin{aligned}
 a &= -1 \quad b = 1 \\
 T &= \frac{1+25}{2} = 1.618 \\
 x_1 &= a + (2-1)(b-a) \\
 x_1 &= -0.236 \\
 x_2 &= b - (2-1)(b-a) \\
 x_2 &= 0.236 \\
 D(x) &= (x-x_0)^2 + (f(x)-y_0)^2 \\
 f(x) &= x^2 + 5 \\
 f(x_1) &= 5.056, f(x_2) = 5.056
 \end{aligned}$$

$$\begin{aligned}
 a) \quad x_0 &= 0 \quad y_0 = 0 \\
 D(x_1) &= (x_1)^2 + (f(x_1))^2 \\
 D(x_1) &= 25.61 \\
 D(x_2) &= 25.619 \\
 \therefore D(x_1) &= D(x_2) \\
 \therefore x &= 0 \quad f(x) = 5
 \end{aligned}$$

$$\begin{aligned}
 b) \quad x_0 &= -4 \quad y_0 = 0 \\
 D(x_1) &= (0.236 + 4)^2 + f(x_1)^2 \\
 D(x_1) &= 39.73 \\
 D(x_2) &= 43.51 \\
 D(x_1) &< D(x_2) \\
 b &= 0.236 \\
 x_2 &= -0.236 \\
 x_1 &= a + (2-1)(b-a) \\
 x_1 &= -0.528 \\
 \text{best } x &= -0.382 \\
 f(x) &= 5.146
 \end{aligned}$$

$$\begin{aligned}
 c) \quad x_0 &= -8 \quad y_0 = 0 \\
 D(x_1) &= (0.236 + 8)^2 + f(x_1)^2 \\
 D(x_1) &= 85.88 \\
 D(x_2) &= 93.39 \\
 D(x_1) &< D(x_2) \\
 b &= 0.236 \\
 x_2 &= -0.236 \\
 x_1 &= a + (2-1)(b-a) \\
 x_1 &= -0.528 \\
 \text{best } x &= -0.382 \\
 f(x) &= 5.146
 \end{aligned}$$

$$\begin{aligned}
 d) \quad x_0 &= 2 \quad y_0 = 0 \\
 D(x_1) &= (0.236 - 2)^2 + f(x_1)^2 \\
 D(x_1) &= 30.56 \\
 D(x_2) &= 28.67 \\
 D(x_1) &> D(x_2) \\
 a &= -0.236 \\
 x_1 &= 0.236 \\
 x_2 &= b - (2-1)(b-a) \\
 x_2 &= 0.528 \\
 \text{best } x &= 0.382 \\
 f(x) &= 5.146
 \end{aligned}$$

$$\begin{aligned}
 e) \quad x_0 &= 6 \quad y_0 = 0 \\
 D(x_1) &= (0.236 - 6)^2 + f(x_1)^2 \\
 D(x_1) &= 64.45 \\
 D(x_2) &= 58.79 \\
 D(x_1) &> D(x_2) \\
 a &= -0.236 \\
 x_1 &= 0.236 \\
 x_2 &= b - (2-1)(b-a) \\
 x_2 &= 0.528 \\
 \text{best } x &= 0.382 \\
 f(x) &= 5.146
 \end{aligned}$$

Part 2

$$(0, 0.5), (2, 3.5), (1, 1.5), (3, 7.5)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

analytical line

$$\begin{aligned}
 &\frac{1}{4} \sum_{i=1}^4 (y - (mx + b))^2 \\
 &= \frac{1}{4} \sum_{i=1}^4 (y^2 - 2ymx + b^2 + m^2x^2 + 2mbx)
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial}{\partial m} \rightarrow 0 &= \frac{1}{4} \sum_{i=1}^4 (-2y + 2mx + 2b) \\
 0 &= \frac{2}{4} (-31 + 14m + 6b) \\
 b &= \frac{31}{6} - \frac{14m}{2}
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial}{\partial b} \rightarrow 0 &= \frac{1}{4} \sum_{i=1}^4 (-2y + 2mx + 2b) \\
 0 &= \frac{2}{4} (-13 + 6m + 4b) \\
 b &= \frac{13}{4} - \frac{6m}{4}
 \end{aligned}$$

$$\begin{aligned}
 \frac{13}{4} - \frac{6m}{4} &= \frac{31}{6} - \frac{14m}{2} \\
 m &= 2.3, b = -0.2 \\
 y &= 2.3x - 0.2
 \end{aligned}$$

$$MSE = \frac{1}{4} \sum_{i=1}^4 (y_i - 2.3x_i + 0.2)^2$$

$$MSE = \frac{1}{4} (0.7^2 + 0.9^2 + 0.6^2 + 0.8^2)$$

$$MSE = 0.575$$

newton raphson line

$$x_1 = x_0 - H^{-1}g$$

$$\begin{aligned}
 f(x) &= \frac{1}{4} \sum_{i=1}^4 (y - mx - b)^2 & \frac{\partial^2 f}{\partial m^2} &= \frac{1}{4} \sum 2x^2 \\
 \frac{\partial f}{\partial m} &= \frac{1}{4} \sum (-2yx + 2mx^2 + 2xb) & \frac{\partial^2 f}{\partial b^2} &= \frac{1}{4} \sum 2 \\
 \frac{\partial f}{\partial b} &= \frac{1}{4} \sum (-2y + 2mx + 2b) & \frac{\partial^2 f}{\partial m \partial b} &= \frac{1}{4} \sum 2x
 \end{aligned}$$

$$g = \begin{bmatrix} \frac{\partial f}{\partial m} \\ \frac{\partial f}{\partial b} \end{bmatrix} \quad H = \begin{bmatrix} \frac{\partial^2 f}{\partial m^2} & \frac{\partial^2 f}{\partial m \partial b} \\ \frac{\partial^2 f}{\partial m \partial b} & \frac{\partial^2 f}{\partial b^2} \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 3 & 2 \end{bmatrix}$$

$$\det = 14 - 9 = 5$$

$$H^{-1} = \frac{1}{5} \begin{bmatrix} 2 & -3 \\ -3 & 7 \end{bmatrix}$$

$$\text{guess } m = 1, b = 1$$

$$g = \begin{bmatrix} \frac{2}{4} (-31 + 14m + 6b) \\ \frac{2}{4} (-13 + 6m + 4b) \end{bmatrix} = \begin{bmatrix} -5.5 \\ -1.5 \end{bmatrix}$$

$$H^{-1}g = \frac{1}{5} \begin{bmatrix} 2 & -3 \\ -3 & 7 \end{bmatrix} \begin{bmatrix} -5.5 \\ -1.5 \end{bmatrix}$$

$$H^{-1}g = \frac{1}{5} \begin{bmatrix} 11 + 4.5 \\ 16.5 - 10.5 \end{bmatrix}$$

$$H^{-1}g = \begin{bmatrix} -1.3 \\ 1.2 \end{bmatrix}$$

$$x_1 = (1, 1) - (-1.3, 1.2)$$

$$x_1 = (2.3, -0.2)$$

analytical parabola

$$\frac{1}{4} \sum_{i=1}^4 (y - (ax^2 + bx + c))^2$$

$$\begin{aligned}
 \frac{\partial}{\partial a} \rightarrow 0 &= \frac{1}{4} \sum_{i=1}^4 (-2yx^2 + 2ax^2 + 2bx + 2cx) \\
 0 &= \frac{2}{4} (-83 + 14a + 36b + 14c)
 \end{aligned}$$

$$\frac{\partial}{\partial b} \rightarrow 0 = \frac{1}{4} \sum_{i=1}^4 (-2yx + 2ax + 2bx^2 + 2cx)$$

$$\textcircled{2} \quad 0 = \frac{2}{4} (-31 + 36a + 14b + 6c)$$

$$\frac{\partial}{\partial c} \rightarrow 0 = \frac{1}{4} \sum_{i=1}^4 (-2y + 2ax + 2bx^2 + 2cx)$$

$$\textcircled{3} \quad 0 = \frac{2}{4} (-13 + 14a + 6b + 4c)$$

$$a = 0.75 \quad b = 0.05 \quad c = 0.55$$

$$y = 0.75x^2 + 0.05x + 0.55$$

$$MSE = \frac{1}{4} \sum_{i=1}^4 (y_i - 0.75x_i^2 - 0.05x_i - 0.55)^2$$

$$MSE = \frac{1}{4} (-0.05^2 - 0.15^2 + 0.15^2 + 0.05^2)$$

$$MSE = 0.0125$$

newton raphson parabola

$$g = \begin{bmatrix} \frac{\partial f}{\partial a} \\ \frac{\partial f}{\partial b} \\ \frac{\partial f}{\partial c} \end{bmatrix}$$

$$H = \frac{1}{2} \begin{bmatrix} \sum x^4 & \sum x^3 & \sum x^2 \\ \sum x^3 & \sum x^2 & \sum x \\ \sum x^2 & \sum x & \sum 1 \end{bmatrix}$$

$$H = \frac{1}{2} \begin{bmatrix} 49 & 18 & 7 \\ 18 & 7 & 3 \\ 7 & 3 & 2 \end{bmatrix}$$

$$H^{-1} = \frac{1}{10} \begin{bmatrix} 5 & -15 & 5 \\ -15 & 49 & -21 \\ 5 & -21 & 19 \end{bmatrix}$$

$$\text{guess } a = 0, b = 0, c = 0$$

$$g = \begin{bmatrix} \frac{2}{4} (-83 + 14a + 36b + 14c) \\ \frac{2}{4} (-31 + 36a + 14b + 6c) \\ \frac{2}{4} (-13 + 14a + 6b + 4c) \end{bmatrix}$$

$$g = \begin{bmatrix} -41.5 \\ -15.5 \\ -6.5 \end{bmatrix}$$

$$H^{-1}g = (-0.75, -0.05, -0.55)$$

$$x_1 = (0, 0, 0) - (-0.75, -0.05, -0.55)$$

$$x_1 = (0.75, 0.05, 0.55)$$